

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

ASSIGNMENT- V

Subject: Software Project Management

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Ques -1 : Change control process handled in software project. Describe the statement.

Ans : In a software project, changes in requirements, design, or implementation are common. To manage these changes in a systematic and controlled manner, a Change Control Process is used.

It ensures that all changes are properly evaluated, approved, and implemented without negatively affecting the project.

Steps in Change Control Process

1. Change Request Submission

- A change request (CR) is submitted by stakeholders such as clients, developers, or testers.
- Example: adding a new feature, modifying an existing function, or fixing a bug.

2. Impact Analysis

- The effect of the proposed change is analyzed in terms of:
 - Cost
 - Time/Schedule
 - Resources
- This step helps in understanding the feasibility of the change.

3. Approval or Rejection

- The Change Control Board (CCB) reviews the request.
- Decision is made whether to:
 - Approve the change
 - Reject the change

4. Implementation

- If approved, the change is implemented by the development team.
- Necessary modifications are made in the system.

5. Testing and Verification

- The implemented change is tested to ensure it works correctly.
- It also ensures no new errors are introduced.

6. Documentation Update

- All related documents (SRS, design documents, etc.) are updated.
- Proper records are maintained for future reference.

Ques-2: Describe the recruitment process for choosing the right person for a job (or) Explain how new staff can be selected and induced into a project.

Ans: Recruitment Process in Software Projects / Selection and Induction of New Staff

Recruitment is the process of selecting the right person for a specific job in a software project. It ensures that skilled and suitable candidates are hired and properly integrated into the project team.

Recruitment & Selection Process

1. Job Analysis

- Identify the job requirements, responsibilities, and skills needed.
- Define roles such as developer, tester, project manager, etc.

2. Job Description & Specification

- Prepare a job description (duties, responsibilities).
- Define qualifications, experience, and skills required.

3. Sourcing Candidates

- Candidates are invited through:
 - Job portals
 - Campus recruitment
 - Internal hiring
- Aim is to attract a large pool of applicants.

4. Screening & Shortlisting

- Applications are reviewed.
- Candidates are shortlisted based on qualifications and experience.

5. Selection Process

- Includes:
 - Written tests (technical/aptitude)
 - Interviews (HR + Technical)
- Best candidate is selected based on performance.

6. Job Offer

- Selected candidate is given an offer letter with salary, role, and terms.

Ques-3: Explain method of staff selection and its merits and demerits.

Ans:- Methods of Staff Selection and Their Merits & Demerits

Staff selection is the process of choosing suitable candidates for a job based on their skills, qualifications, and project requirements. Different methods are used to ensure the right person is selected for the right role

Methods of Staff Selection

1. Interview Method

- Candidates are assessed through face-to-face or virtual interviews.
- Includes HR and technical interviews.

Merits:

- Direct interaction with candidate
- Helps judge communication and confidence
- Flexible and widely used

Demerits:

- Subjective and may be biased
- Time-consuming
- Depends on interviewer's skill

2. Written Test

- Includes aptitude tests, technical tests, or coding tests.

Merits:

- Objective evaluation
- Easy to compare candidates
- Tests technical knowledge

Demerits:

- Does not assess practical skills fully
- Some candidates may perform poorly under pressure

3. Group Discussion (GD)

- Candidates discuss a topic in a group.

Merits:

- Evaluates communication skills
- Tests leadership and teamwork
- Observes behavior in group situations

Demerits:

- Dominant candidates may overshadow others
- Not suitable for all job roles

4. Psychometric Test

- Measures personality, attitude, and behavior.

Merits:

- Helps understand candidate's mindset
- Useful for team compatibility

Demerits:

- Results may not always be accurate
- Can be difficult to interpret

5. Assessment Center Method

- Combination of tests, interviews, and exercises.

Merits:

- Comprehensive evaluation

- More reliable results

Demerits:

- Expensive
- Time-consuming

6. Reference Check

- Verifying candidate's past experience and background.

Merits:

- Provides reliable information
- Reduces hiring risk

Demerits:

- Information may be biased
- Not always accurate

Ques-4: Summarize software configuration management. Write down its various configuration items in detail.

Ans: Software Configuration Management (SCM) & Its Configuration Items

Software Configuration Management (SCM) is the process of managing, organizing, and controlling changes in software throughout its life cycle.

It ensures that the software system remains consistent, traceable, and reliable even after multiple modifications.

Objectives of SCM

- Control and manage changes
- Maintain version history
- Ensure consistency of software
- Improve coordination among team members

Activities of SCM

- Configuration Identification
- Version Control
- Change Control
- Configuration Status Accounting
- Configuration Audit

Configuration Items (CIs)

Configuration Items are the components of a software system that are managed under SCM.

1. Source Code

- All program files written by developers
- Example: .c, .java, .py files
- Most critical CI

2. Documentation

- Includes all project-related documents:
 - SRS (Software Requirement Specification)

- Design documents
- User manuals
- Helps in understanding and maintaining the system

3. Executable Files

- Compiled or built versions of the software
- Example: .exe, .jar
- Used for deployment and testing

4. Test Cases & Test Data

- Test scripts, test plans, and datasets
- Ensures software quality and validation

5. Configuration Files

- Files that define system settings
- Example: .xml, .json, .env
- Used to control application behavior

6. Libraries & External Components

- Third-party libraries and APIs used in the project
- Important for dependency management

7. Build & Deployment Scripts

- Scripts used for compiling, building, and deploying software
- Automates development process

8. Change Requests & Reports

- Records of requested and approved changes
- Helps in tracking modifications

Importance of Configuration Items

- Helps in version control
- Makes debugging easier
- Ensures consistency
- Supports teamwork and collaboration

Ques-5: Illustrate Case tools and their utilization in software development

Ans: CASE (Computer-Aided Software Engineering) tools are software applications that support and automate various activities of the software development life cycle (SDLC).

They help developers in designing, coding, testing, and maintaining software efficiently.

Types of CASE Tools

1. Upper CASE Tools

- Used in early stages of development (planning, analysis, design)
- Examples:
 - Requirement analysis tools
 - UML diagram tools

Help in system modeling and documentation

2. Lower CASE Tools

- Used in later stages (coding, testing, maintenance)
- Examples:
 - Code generators
 - Testing tools

Help in implementation and debugging

3. Integrated CASE Tools (I-CASE)

- Support the entire SDLC
 - Combine both Upper and Lower CASE tools

Utilization of CASE Tools in SDLC

1. Requirement Analysis

- Helps in gathering and documenting requirements
- Improves clarity and reduces errors

2. System Design

- Used to create diagrams like:
 - UML diagrams
 - Data flow diagrams (DFD)
 - Makes system structure easy to understand

3. Coding

- Some tools generate code automatically
- Reduces manual effort and errors

4. Testing

- Automated testing tools help in:
 - Unit testing
 - Integration testing
 - Improves software quality

5. Maintenance

- Helps in tracking changes and updating software
- Supports version control

Advantages of CASE Tools

- Increases productivity
- Improves software quality
- Reduces development time
- Provides better documentation
- Enhances team collaboration

Disadvantages of CASE Tools

- High cost
- Requires training
- Not flexible for all projects
- Dependency on tools

Ques 6: What do you understand by good decision making? Also mention obstacles to good group decision making. Give measure to reduce obstacles of group decision making.

Ans: - Good Decision Making, Obstacles in Group Decision Making, and Measures

Good Decision Making

Good decision making is the process of selecting the best course of action from available alternatives based on proper analysis, facts, and objectives. A good decision is:

- Logical and rational
- Based on accurate information
- Focused on achieving project goals
- Aware of risks and consequences

Obstacles to Good Group Decision Making

1. Groupthink

Members avoid expressing different opinions to maintain harmony, which leads to poor decisions.

2. Dominance by Individuals

One or a few members dominate the discussion, preventing others from contributing.

3. Lack of Communication

Poor communication leads to misunderstandings and loss of important information.

4. Conflict Among Members

Personal conflicts distract from the actual problem and reduce decision quality.

5. Lack of Information

Decisions are made without sufficient or accurate data.

6. Time Pressure

Limited time leads to rushed decisions without proper analysis.

7. Large Group Size

Difficult to manage and slows down the decision-making process.

Measures to Reduce Obstacles

1. Encourage Open Communication

Create an environment where all members can share ideas freely.

2. Avoid Groupthink

Encourage critical thinking and consideration of alternative solutions.

3. Balanced Participation

Ensure equal involvement of all members and limit dominance.

4. Conflict Management

Resolve conflicts professionally and keep focus on the problem.

5. Use Proper Data

Collect and analyze relevant information before making decisions.

6. Effective Leadership

A good leader guides discussion and maintains focus.

7. Proper Time Management

Allocate enough time for discussion and avoid rushed decisions.

Ques-7: Explain the various Motivation theories formulated for motivating people to work.

Ans: Motivation is the process of encouraging people to work efficiently and achieve organizational goals. In software project management, motivation is important to improve productivity, job satisfaction, and team performance. Various theories have been proposed to explain how people can be motivated.

1. Maslow's Hierarchy of Needs

Proposed by Abraham Maslow, this theory states that human needs are arranged in a hierarchy:

1. Physiological Needs – food, salary, basic needs
2. Safety Needs – job security, safe environment
3. Social Needs – teamwork, relationships
4. Esteem Needs – recognition, respect
5. Self-Actualization – personal growth, creativity

Employees are motivated to satisfy lower-level needs before moving to higher-level needs.

2. Herzberg's Two-Factor Theory

Proposed by Frederick Herzberg, it divides factors into two categories:

- Hygiene Factors (dissatisfiers): salary, job security, working conditions
 - Their absence causes dissatisfaction, but presence does not strongly motivate
- Motivators (satisfiers): achievement, recognition, responsibility
 - These increase job satisfaction and motivation

3. McGregor's Theory X and Theory Y

Theory X

- Assumes employees are lazy and avoid work
- Require strict supervision and control

Theory Y

- Assumes employees are responsible and self-motivated
- Encourage participation and creativity

Modern management prefers Theory Y.

4. McClelland's Need Theory

Proposed by David McClelland, it focuses on three needs:

- Need for Achievement (nAch): desire to succeed
- Need for Power (nPow): desire to control or influence
- Need for Affiliation (nAff): desire for relationships

Different people are motivated by different needs.

5. Expectancy Theory

Proposed by Vroom, it states that motivation depends on:

- Expectancy: effort will lead to performance
- Instrumentality: performance will lead to reward
- Valence: value of the reward

Employees work harder when they believe their effort will be rewarded.

6. Equity Theory

This theory states that employees compare their input-output ratio with others.

- If they feel treated fairly → motivated
- If they feel unfair → demotivated

Ques-8: Describe the following:

1. Oldham Hackman Job Characteristics

2. Maslow's Hierarchy of needs

Ans:- 1. Hackman–Oldham Job Characteristics Model

It explains how job design affects motivation. It includes five core characteristics:

- Skill Variety – Use of different skills
- Task Identity – Completing a whole task
- Task Significance – Impact of work on others

- Autonomy – Freedom in work
- Feedback – Information about performance

2. Maslow's Hierarchy of Needs

Proposed by Abraham Maslow, it states that human needs are arranged in levels:

1. Physiological – basic needs (food, salary)
2. Safety – security, job safety
3. Social – relationships, teamwork
4. Esteem – respect, recognition
5. Self-Actualization – personal growth

